Policy Improvement BSDE: Fixed-Policy Evaluation in a Toy Consumption-Saving Model and a Three-Country International Macro Model

Quantitative Macro Research Note

Abstract

This technical note develops a block-wise Backward Stochastic Differential Equation (BSDE) evaluation framework for continuous-time stochastic general equilibrium models. The core principle is that BSDEs can only reliably approximate value functions or asset price functions when all policy rules and state drift dynamics are held fixed; policy or equilibrium rate updates are treated as outer block iterations separated from inner BSDE training. We first present a canonical two-state stochastic neoclassical growth toy model with Ornstein–Uhlenbeck TFP shocks, solved via BSDE-based Howard policy iteration. We then extend the block BSDE method to a three-country heterogeneous-agent international macro model with expert capital owners and household savers, where we alternate BSDE asset price evaluation and global risk-free rate updating via block Gauss–Seidel iteration. We provide full model dynamics, BSDE residual loss formulations, iterative algorithms, closed-form equilibrium restrictions, and analytical comparative static results for structural parameters. Benchmark comparisons against standard HJB finite difference solvers and cross-parameter sensitivity experiments are discussed.

1 Toy Continuous-Time Stochastic Consumption-Saving Model

This section introduces a minimal representative-agent stochastic growth environment to illustrate the core BSDE policy iteration paradigm: freeze the consumption policy, train BSDE critic networks for value V and martingale diffusion Z, then update consumption via Hamiltonian first-order conditions (FOC).

1.1 Model Primitives and State Dynamics

The model features two Markov state variables: private capital stock k and stationary aggregate productivity shock z. Instantaneous output is $y = e^z k^\alpha$. Capital accumulation follows

$$dk = [e^z k^\alpha - \delta k - c(k,z)]\, dt,$$

where $c(k,z)$ denotes instantaneous consumption flow. Productivity evolves as a mean-reverting Ornstein–Uhlenbeck (OU) process driven by a standard Brownian motion W_t:

$$dz = \kappa_z\,(\bar{z} - z)\, dt + \sigma_z dW_t.$$

The household maximizes infinite-horizon discounted CRRA utility:

$$V(k,z) = \sup_{\{c_t\}} \mathbb{E}\left[\int_0^\infty e^{-\rho t} u(c_t) dt \,\middle|\, k_0 = k,\ z_0 = z\right],$$

with utility kernel

$$u(c) = \frac{c^{1-\gamma} - 1}{1-\gamma}, \quad \gamma > 0, \gamma \neq 1.$$

1.2 Baseline Calibration

Table 1: Toy Model Structural Parameters

Parameter	Description	Value
α	Capital income share	0.36
δ	Capital depreciation rate	0.08
γ	Coefficient of relative risk aversion	2.00
ρ	Subjective time discount rate	0.05
κ_z	TFP mean reversion speed	0.30
$\bar{z}$	Long-run mean TFP shock	0
σ_z	TFP diffusion volatility	0.023
Δt	Discrete Euler BSDE time step	0.005

Numerical state grid bounds for training and interpolation:

$$k \in [0.05,\ 30], \qquad z \in [-0.12,\ 0.12].$$

1.3 Fixed-Policy BSDE Formulation

Conditional on a frozen consumption policy $c(k, z)$, the value process V_t solves the backward SDE

$$dV_t = -\left[u(c_t) - \rho V_t\right] dt + Z_t dW_t,$$

where Z_t is the BSDE martingale diffusion term capturing exposure to Brownian productivity shocks. Discretize forward state transition one step to future state $S_{t+\Delta t} = (k', z')$. Define the one-step BSDE residual

$$R_t = V_t + \left[\rho V_t - u(c_t)\right] \Delta t + Z_t \Delta W_t - V(S_{t+\Delta t}),$$

where $\Delta W_t \sim \mathcal{N}(0, \Delta t)$ is the discrete Brownian increment. The critic network training loss minimizes the expected squared residual:

$$\mathcal{L}_{\text{BSDE}} = \mathbb{E}\left[R_t^2\right].$$

Crucially, consumption c is held fixed throughout inner BSDE loss minimization; the network only approximates V and Z given the pre-specified policy.

1.4 BSDE Policy Iteration Algorithm (Howard Iteration)

The algorithm separates policy evaluation and policy improvement into disjoint outer rounds, with no joint optimization over consumption, value, and diffusion within a single training loop:

(1) Initialize a feasible starting consumption guess: $c^0(k, z) = 0.1 \cdot e^z k^\alpha$.

(2) Policy Evaluation (Inner BSDE Step): Freeze policy c^n. Train a joint feedforward critic network mapping $(k, z) \mapsto (V^n, Z^n)$ by minimizing $\mathcal{L}_{\text{BSDE}}$ over batches of simulated state trajectories.

(3) Policy Improvement (Outer Update Step): Compute the marginal value of capital $V_k^n = \partial V^n / \partial k$. Apply the Hamiltonian FOC $u'(c) = V_k$ to solve for updated consumption:

$$c^{n+1} = (V_k^n)^{-1/\gamma}.$$

Numerical safeguards (damped updates, consumption lower bounds, feasibility clipping to ensure positive investment) are applied to avoid policy explosion near grid boundaries. The updated policy c^{n+1} is stored as a grid-interpolated object for the next iteration.

(4) Iterate for fixed total rounds (baseline experiment: 20 outer iterations, 20,000 BSDE training steps per evaluation).

The iterative mapping is compactly written as

$$c^n \mapsto \left(V^{c^n}, Z^{c^n}\right) \mapsto c^{n+1}.$$

At the theoretical fixed point $c^* = \mathcal{G}(V^{c^*})$, the policy is self-consistent with its associated value function. However, neural approximation error, inaccurate state derivatives near grid edges, and non-contractive outer update maps break standard theoretical convergence guarantees for exact Howard iteration; the continuous-time HJB finite difference solver serves as a benchmark validation tool.

1.5 Numerical Output Pipeline

After each outer policy iteration round, the implementation exports:

(i) Slices at steady-state productivity $z = 0$: plots of $V(k, 0)$, $c(k, 0)$, marginal value $V_k(k, 0)$, BSDE diffusion $Z(k, 0)$ (saved as `toy_bsde_round_XX.png`).

(ii) Convergence charts tracking cross-grid maximum absolute deviations of V and c across all 20 rounds.

(iii) Benchmark overlay figure `toy_hjb_bsde_overlay.png` comparing BSDE policy iteration against standard HJB finite difference Howard iteration.

(iv) Full grid numerical dataset serialized as compressed `.npz` files for post-simulation quantitative analysis.

2 Three-Country International Macro Model

We extend the fixed-drift BSDE evaluation logic to a multi-country open-economy framework with two agent types: capital-owning experts and wealth-saving households. The core block iteration structure is analogous to the toy model—fix a subset of equilibrium objects, run BSDE evaluation, then update the remaining equilibrium price block—though the update step here targets a global risk-free rate rather than a household consumption policy.

2.1 Economic Environment

Three symmetric countries indexed $i \in \{1, 2, 3\}$ with segmented physical capital markets:

- Expert agents hold all domestic productive capital; cross-border direct capital investment is prohibited.

- Households trade a single common international risk-free bond, equalizing the global equilibrium interest rate $r(S)$ across all regions.
- Household preferences are logarithmic, implying aggregate consumption is proportional to total wealth with constant consumption rate ρ.

The aggregate state vector S collects wealth distribution shares:

$$S = (\eta_1, \eta_2, \eta_3, \zeta_1, \zeta_2), \qquad \zeta_3 = 1 - \zeta_1 - \zeta_2,$$

where η_i = expert wealth share in country i, ζ_i = country i's share of total world household wealth.

2.2 Capital, Investment, and Asset Price Dynamics

Let q_i denote the market price of installed physical capital in country i. Linear investment adjustment costs produce investment flow

$$\iota_i = \frac{q_i - 1}{\psi}.$$

Country-i capital stock evolves with constant fundamental volatility σ:

$$\frac{dK_i}{K_i} = (a - \iota_i)\, dt + \sigma dW_i.$$

The capital price follows a diffusion process with drift $\mu_{q,i}$ and volatility matrix σ_q (row $i = \sigma_{q,i\cdot}$):

$$\frac{dq_i}{q_i} = \mu_{q,i} dt + \sigma_{q,i\cdot} d\boldsymbol{W}.$$

Define total capital return exposure for country i:

$$\sigma^K_{i\cdot} = \sigma_{q,i\cdot} + \sigma \boldsymbol{e}_i^\top,$$

where $\boldsymbol{e}_i$ is the unit vector selecting the i-th Brownian shock.

2.3 Expert Euler Equation and Price Drift Restriction

Conditional on a fixed global rate function $r(S)$, the expert's asset pricing Euler condition delivers a closed-form expression for the price drift $\mu_{q,i}$ used as the BSDE drift input:

$$\mu_{q,i} = -\frac{1 + a\psi}{\psi q_i} - \frac{\log q_i}{\psi} + \left(\frac{1}{\psi} + \delta\right) - \sigma(\sigma_q)_{ii} + \frac{\|\sigma^K_{i\cdot}\|^2}{\eta_i} + r.$$

The quadratic risk premium term driving financial amplification is

$$\mathrm{RP}_i = \frac{\|\sigma^K_{i\cdot}\|^2}{\eta_i}.$$

Risk compensation rises nonlinearly as expert wealth share η_i shrinks, generating powerful balance-sheet amplification of fundamental shocks.

2.4 State Laws of Motion for Wealth Shares

2.4.1 Expert wealth share η_i

$$d\eta_i = \left[\left(\frac{1 + a\psi}{\psi q_i} - \frac{1}{\psi} - \rho - \chi\right)\eta_i + \left(\frac{1}{\eta_i} - 1\right)^2 \eta_i \|\sigma^K_{i\cdot}\|^2\right] dt + (1 - \eta_i)\sigma^K_{i\cdot} d\boldsymbol{W}.$$

2.4.2 Household wealth share ζ_i

Define country-i household capital return drift μ_i^K and world portfolio aggregates:

$$\mu_i^K = -\frac{1 + a\psi}{\psi q_i} + \frac{1}{\psi} + \frac{\|\sigma_{i\cdot}^K\|^2}{\eta_i} + r, \quad \mu_H = \sum_j \zeta_j \mu_j^K, \quad \sigma_H = \sum_j \zeta_j \sigma_{j\cdot}^K.$$

The wealth share diffusion is

$$\frac{d\zeta_i}{\zeta_i} = \left[\mu_i^K - \mu_H - \sigma_H \left(\sigma_{i\cdot}^K - \sigma_H\right)^\top\right] dt + \left(\sigma_{i\cdot}^K - \sigma_H\right) d\boldsymbol{W}.$$

2.5 Market Clearing Condition

Final goods market clearing imposes an exact cross-country restriction on capital prices, enforced directly via network output transformation (no soft loss penalty):

$$\sum_{i=1}^{3} \frac{\zeta_i}{q_i} = \frac{1 + \psi\rho}{1 + a\psi}.$$

2.6 Three-Country Block BSDE Iteration Algorithm

The critical identification principle: BSDE evaluation is well-conditioned only when all drift coefficients and state transition laws are fixed; simultaneous optimization over asset prices and the global interest rate destabilizes training. We implement a nonlinear block Gauss–Seidel alternating minimization scheme with two core steps per outer iteration:

2.6.1 Step A: Fixed $r(S)$ — BSDE Evaluation of Capital Prices $q_i, \sigma_{q,i\cdot}$

Given an initial guess for the interest rate function (baseline $r(S) = 0.03$), simulate forward state trajectories $S^{+,d}$ over independent Brownian shock batches. Regress future capital prices on current states and shock increments to recover BSDE intercept and slope terms:

$$q_i(S^{+,d}) = \alpha_i(S) + \beta_{i\cdot}(S)\Delta\boldsymbol{W}^d + \varepsilon_i^d.$$

BSDE consistency requires

$$\alpha_i = q_i + q_i\mu_{q,i}\Delta t, \qquad \beta_{i\cdot} = q_i\sigma_{q,i\cdot}.$$

The normalized training loss for the price-volatility network is

$$\mathcal{L}_q = \frac{\left\|\alpha - q - q\mu_q\Delta t\right\|^2}{\mathbb{E}[\mu_q^2]} + \frac{\left\|\beta/q - \sigma_q\right\|^2}{\frac{1}{2}\mathbb{E}\left[\|\sigma_q\|^2 + \|\beta/q\|^2\right]},$$

with detached normalization constants to prevent trivial scale-based loss reduction.

2.6.2 Step B: Fixed (q, σ_q) — Update Global Risk-Free Rate $r(S)$

Freeze the trained price-volatility network. Train a separate neural network for $r(S)$ against the identical BSDE conditional regression loss, solving for a single common interest rate consistent with all three countries' Euler drift restrictions.

The full outer block iteration mapping is

$$r^n \mapsto \left(q^{n+1}, \sigma_q^{n+1}\right) \mapsto r^{n+1}.$$

This alternating residual minimization shares the "freeze one block, evaluate the other" architecture of the toy model policy iteration, but it is not a household consumption policy improvement step and lacks universal global convergence guarantees.

2.7 Structural Parameter Comparative Static Results

2.7.1 Varying production efficiency $a \in \{0.04, 0.07, 0.10\}$

Fix fundamental volatility $\sigma = 0.023$. Let an equilibrium triple $(q_i^0, \sigma_{q,i\cdot}^0, r^0)$ correspond to efficiency a_0, and define scaling factor

$$\lambda = \frac{1 + a_1\psi}{1 + a_0\psi}.$$

Construct transformed equilibrium objects for alternative a_1:

$$q_i^1 = \lambda q_i^0, \quad \sigma_{q,i\cdot}^1 = \sigma_{q,i\cdot}^0, \quad r^1 = r^0 + \frac{\log\lambda}{\psi}.$$

Investment and capital growth satisfy

$$a_1 - \iota_i^1 = \lambda(a_0 - \iota_i^0) \implies \frac{a_1 - \iota_i^1}{q_i^1} = \frac{a_0 - \iota_i^0}{q_i^0}.$$

All state transition dynamics remain unchanged, and the price diffusion process satisfies

$$\frac{dq_i^1}{q_i^1} = \frac{dq_i^0}{q_i^0} \implies \boxed{\sigma_q^{a_1}(S) = \sigma_q^{a_0}(S)}.$$

Key implication: Shifts in production efficiency a scale the level of capital prices and the global risk-free rate, but leave relative price volatility σ_q unchanged. The absolute volatility exposure $q_i\sigma_{q,i\cdot}$ scales proportionally with q_i.

2.7.2 Varying fundamental shock volatility $\sigma \in \{0.015, 0.023, 0.035\}$

Fix $a = 0.10$; the baseline benchmark is $\sigma = 0.023$. Unlike a, fundamental volatility cannot be absorbed via uniform price and rate rescaling. It directly enters both state diffusion and the quadratic risk premium term:

$$d\eta_i \supset (1 - \eta_i)\big(\sigma_{q,i\cdot} + \sigma \boldsymbol{e}_i^\top\big)d\boldsymbol{W}, \quad \mathrm{RP}_i = \frac{\big\|\sigma_{q,i\cdot} + \sigma \boldsymbol{e}_i^\top\big\|^2}{\eta_i}.$$

Low (0.015) and high (0.035) volatility experiments are warm-started from the converged $\sigma = 0.023$ checkpoint; full output figures are generated only after complete outer iteration convergence.

2.8 Simulation Outputs for Parameter Sensitivity

For each (a, σ) calibration, the code produces two core visualization panels:

(i) 9-panel grid plotting every element of the capital price volatility matrix σ_q.

(ii) 4-by-4 summary grid reporting country capital prices q_i, global risk-free rate r, Euler risk premia RP_i, and total capital return volatility $\|\sigma_{i\cdot}^K\|$.

3 Unifying Discussion of Block BSDE Evaluation

Both the toy consumption-saving model and three-country open-economy framework rely on the same foundational BSDE principle: backward SDEs deliver stable function approximation only when all forward drift dynamics and control/equilibrium price blocks are held fixed during training. The two models differ only in the outer block update step:

- Toy closed-economy model: Outer step = policy improvement via household consumption FOC (Howard policy iteration).
- Three-country open-economy model: Outer step = global risk-free rate adjustment via block Gauss–Seidel, matching cross-country asset pricing Euler equations.

This fixed-drift BSDE paradigm avoids explicit finite-difference PDE discretization of high-dimensional state spaces, replacing grid interpolation with stochastic trajectory sampling and neural function approximation. The method generalizes naturally to richer environments including heterogeneous agents, recursive Epstein-Zin preferences, long-run risk, and multi-asset international portfolios.